

PSET 2 SCREENSHOTS

FATIMA IRFAN 29294

Q1:

```
forphan user/ dorphan user/ attack user/ secret
meta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1
953 total 2000
ballocc: first 988 blocks have been allocated
ballocc: write bitmap block at sector 46
*** Now run 'gdb' in another window.
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp
3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,f
ormat=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0 -S -gdb
tcp::26000

xv6 kernel is booting

hart 1 starting
hart 2 starting

> 137 num = p->trapframe->a7
138 if num < 0 || num > NELEM syscalls || syscalls[num] {
139 // Use num to lookup the system call function for num, call it,
140 // and store its return value in p->trapframe->a0
141 p->trapframe->a0 = syscalls[num]();
142 } else {
remote Thread 1.3 (src) In: syscall L137 PC: 0x80001cfa
s0 = 0x3fffffdcb0, s1 = 0x80007cd0, s2 = 0x800078a0, s3 = 0x0,
s4 = 0x80012de0, s5 = 0x3, s6 = 0x80018970, s7 = 0x0, s8 = 0x80018a90,
s9 = 0x100, s10 = 0xb0, s11 = 0x2, ofile = {0x0 <repeats 16 times>},
--Type <RET> for more, q to quit, c to continue without paging--
cud = 0x80015de0, name = {0x69, 0x6e, 0x69, 0x74, 0x0 <repeats 12 times>}}
(gdb) p /x p->trapframe->a7
$2 = 0xf
(gdb)

*** Now run 'gdb' in another window.
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xv6 kernel is booting

hart 1 starting
hart 2 starting

; /'s/d' > kernel/kernel.sys
make[1]: Leaving directory '/home/ubuntu-os/xv6-labs-2025'
== Test answers-syscall.txt ==
answers-syscall.txt: OK
== Test sandbox_mask ==
$ make qemu-gdb
(gdb) layout asm
Detaching from program: /home/ubuntu-os/xv6-labs-2025/kernel/kernel, process 1
Ending remote debugging.
[Inferior 1 (process 1) detached]
ubuntu-os@ubuntu-os-VirtualBox:~/xv6-labs-2025$ kill -f qemu-system-riscv64
ubuntu-os@ubuntu-os-VirtualBox:~/xv6-labs-2025$ nano answers-syscall.txt
ubuntu-os@ubuntu-os-VirtualBox:~/xv6-labs-2025$
```

Q2:

```
xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ sandbox 32768 README grep xv6 x
grep: cannot open x
$ sandbox 32768 - cat README
cat: cannot open README
$ ./grade-lab-syscall sandbox_mask
exec ./grade-lab-syscall failed
$ QEMU: Terminated

ubuntu-os@ubuntu-os-VirtualBox:~/xv6-labs-2025$ ./grade-lab-syscall sandbox_mask
make: 'kernel/kernel' is up to date.
== Test sandbox_mask == sandbox_mask: OK (2.3s)
(Old xv6.out.sandbox_mask failure log removed)
```

Q3:

```
xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ $ sandbox 32768 README grep xv6 README
# expected: contents of README related lines printed
exec $ failed
$ exec # failed
$ sandbox 32768 README grep xv6 README
xv6 is a re-implementation of Dennis Ritchie's and Ken Thompson's Unix
Version 6 (v6).  xv6 loosely follows the structure and style of v6,
xv6 is inspired by John Lions's Commentary on UNIX 6th Edition (Peer
(kaashoek,rtm@mit.edu).  The main purpose of xv6 is as a teaching
$ sandbox 32768 README grep xv6 x
grep: cannot open x
```

Q4:

```
xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ secret abc123
$ attack
N
$ attack
abc123
$
```

MAKE GRADE:

```
== Test answers-syscall.txt ==
answers-syscall.txt: OK
== Test sandbox_mask ==
$ make qemu-gdb
sandbox_mask: OK (10.3s)
== Test sandbox_fork ==
$ make qemu-gdb
sandbox_fork: OK (1.3s)
== Test sandbox_path ==
$ make qemu-gdb
sandbox_path: OK (3.2s)
== Test sandbox_most ==
$ make qemu-gdb
sandbox_most: OK (1.4s)
== Test sandbox_minus ==
$ make qemu-gdb
sandbox_minus: OK (1.9s)
== Test attack ==
$ make qemu-gdb
attack: OK (1.5s)
      (Old xv6.out.attack failure log removed)
== Test time ==
```